Strategically robust decision-making

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AI and autonomous systems are increasingly decision makers in the wild



nature communications

Article

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A matched case-control analysis of autonomous vs human-driven vehicle accidents

mous' versus Human-Driven Vehicles' accidents. The analysis suggests that accidents of vehicles equipped with Advanced Driving Systems generally have a lower chance of occurring than Human-Driven Vehicles in most of the similar accident scenarios. However, accidents involving Advanced Driving Systems occur more frequently than Human-Driven Vehicle accidents under dawn/dusk or turning conditions, which is 5.25 and 1.98 times higher, respectively.

scenarios are a considerable challenge for AVs such as unprotected left turns at intersections[59]. These situations are complicated by factors like limited priority and variation in trajectories[60]. AVs tend to be overcautious (such as having a longer startup delay during the turning at intersections)[61], which can lead to rear-end or sideswipe accidents with HDVs[62]. Furthermore, multi-interactions caused by mixed flows aggravate uncertainties in detection, such as the superposition of distance and angle measurement error[63]. Conversely, HDVs can adapt and modify their speed more seamlessly than AVs, highlighting the limitations of AVs in comparison to the adeptness of experienced drivers[64]. And AVs face difficulties with executing lane changes or turning in heavy traffic and lack psychological insight[65,66]. In addition, HDVs can predict pedestrian movements and exercise caution based on their driving experience, whereas AVs may struggle with recognizing pedestrians' intentions, potentially leading to emergency braking or accidents due to a lack of understanding of social cues and psychological reasoning[65,67,68].

AI is trained “in silico”



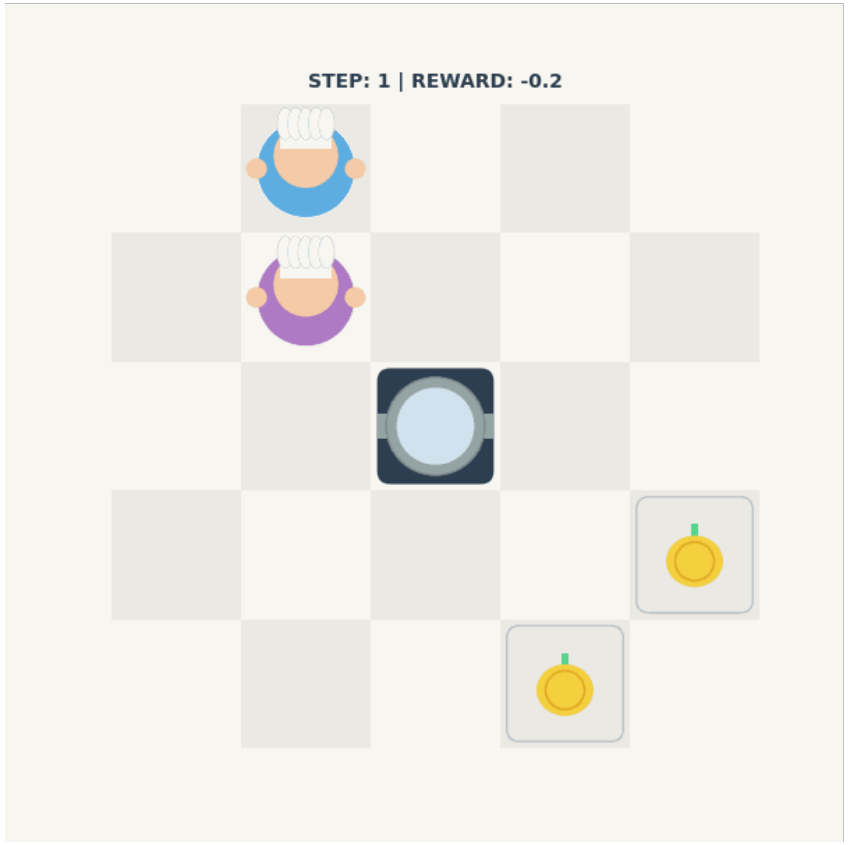
... but deployed “in the wild”, in settings that are

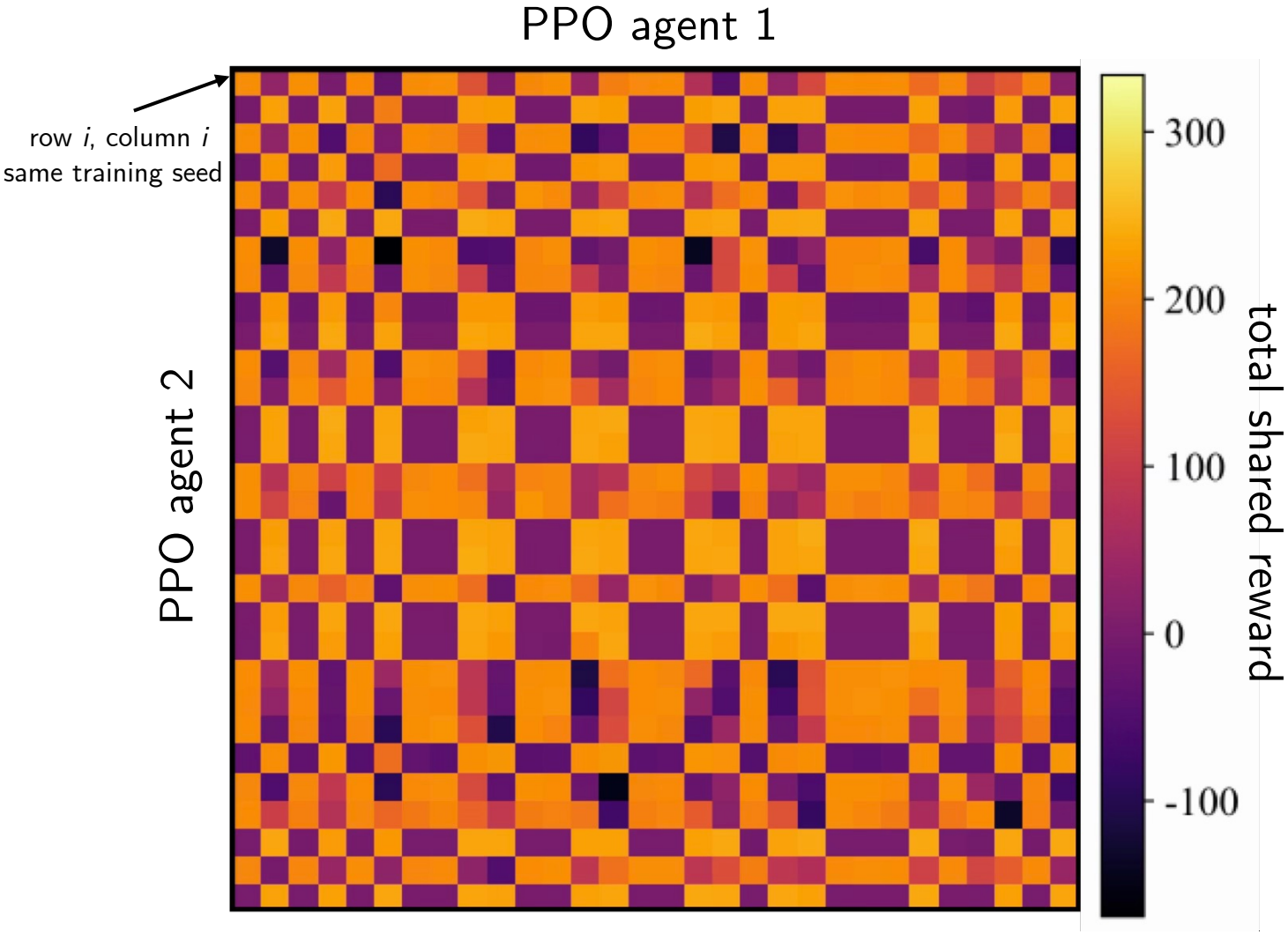
🤖🧍 strategic

🧙 uncertain

🏎️ dynamic

not due to the complexities of the real world





agents learn to free-ride

shared reward for picking up an onion:**+1**
shared reward for putting onion in pot: **+10**
shared cost for collision: **2**
individual cost for effort **0.2/per move**

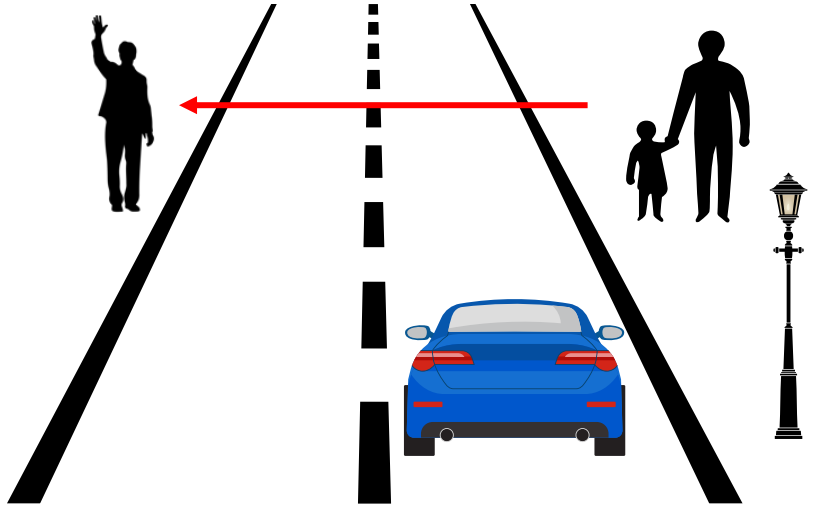
agents trained with PPO, the state-of-art of MARL

strategically robust game theory

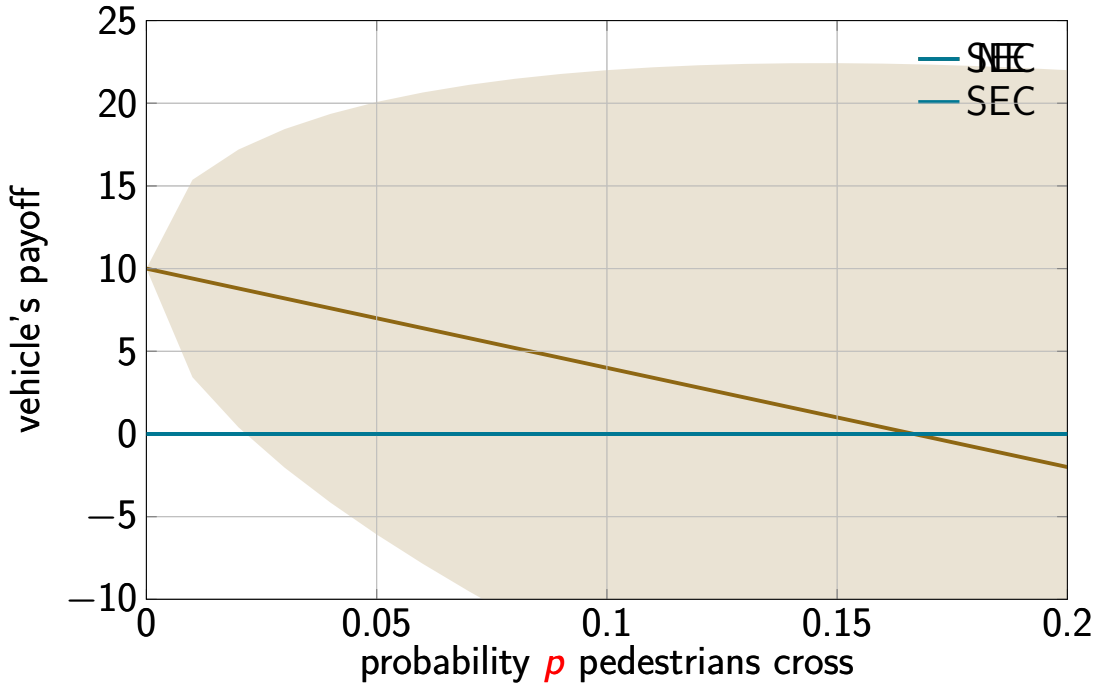
with S. Fricker, S. Bolognani, F. Dörfler, and D. Paccagnan



agents are uncertain about each other



	stop	cross
maintain	(10, –1)	(–50, –100)
decelerate	(9, –1)	(–5, –10)
stop	(0, –1)	(0, 10)



players $i = 1, \dots, N$

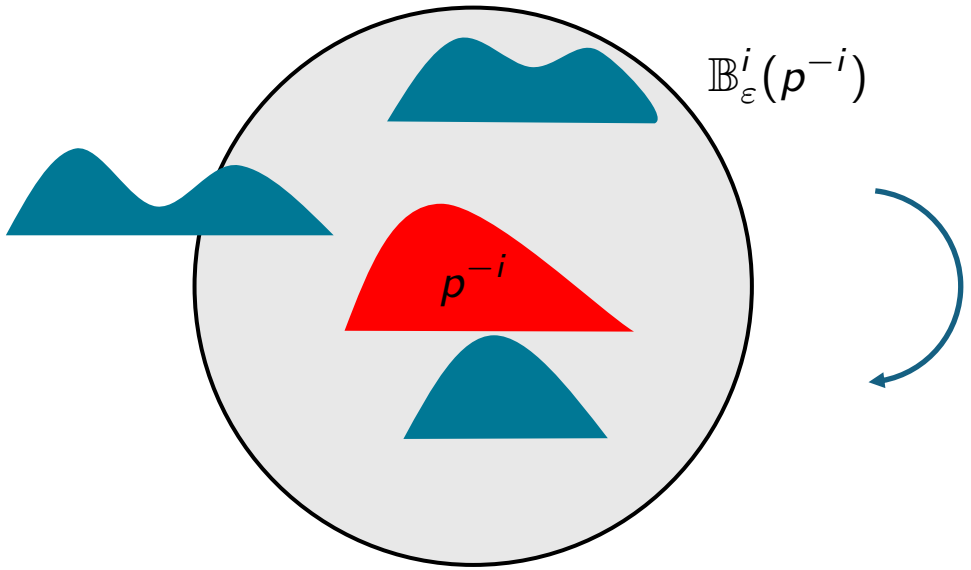
action space $\mathcal{A} = \mathcal{A}_1 \times \dots \times \mathcal{A}_N$

strategy space $\Delta^i = \mathcal{P}^i(\mathcal{A}^i)$

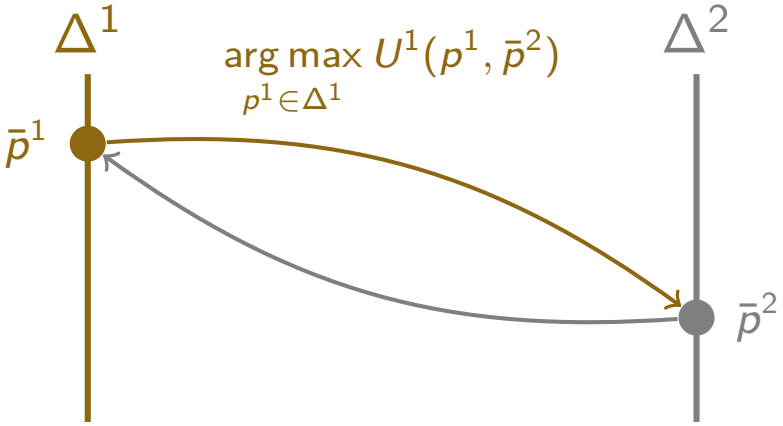
expected utility $U^i(p^i, p^{-i}) = \mathbb{E}^{a^i \sim p^i, a^{-i} \sim p^{-i}}[u^i(a^i, a^{-i})]$

strategies of all players but i

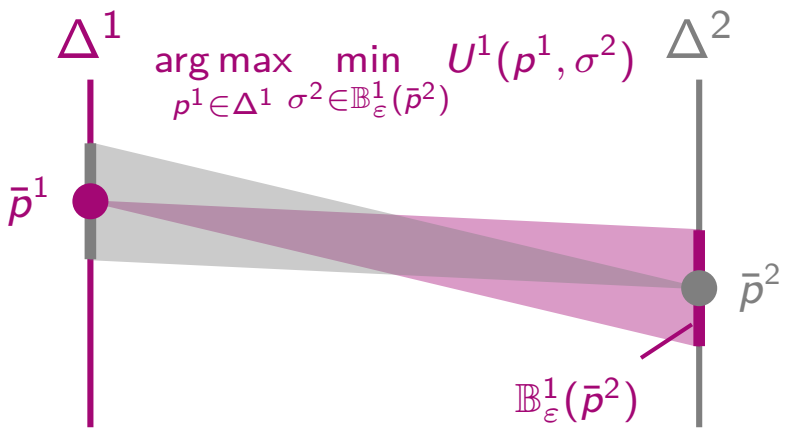
ambiguity sets $\mathbb{B}^i_\varepsilon(p^{-i}) \subseteq \Delta^{-i}$



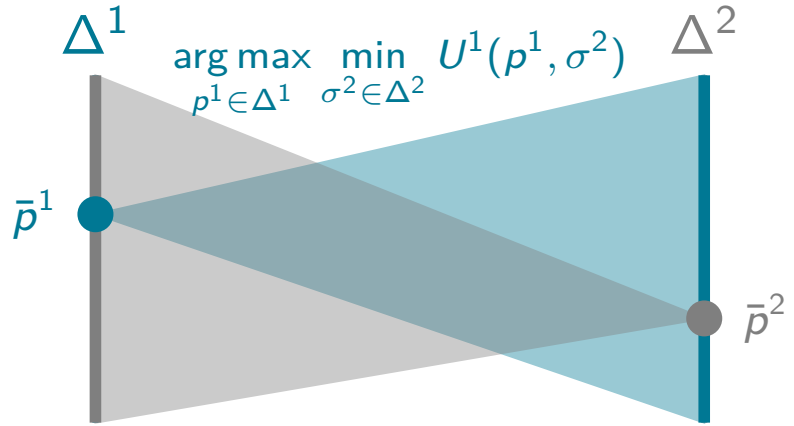
distributionally robust best response



Nash equilibria
i.e., SRE with $\varepsilon = 0$



strategically robust equilibria
(SRE)



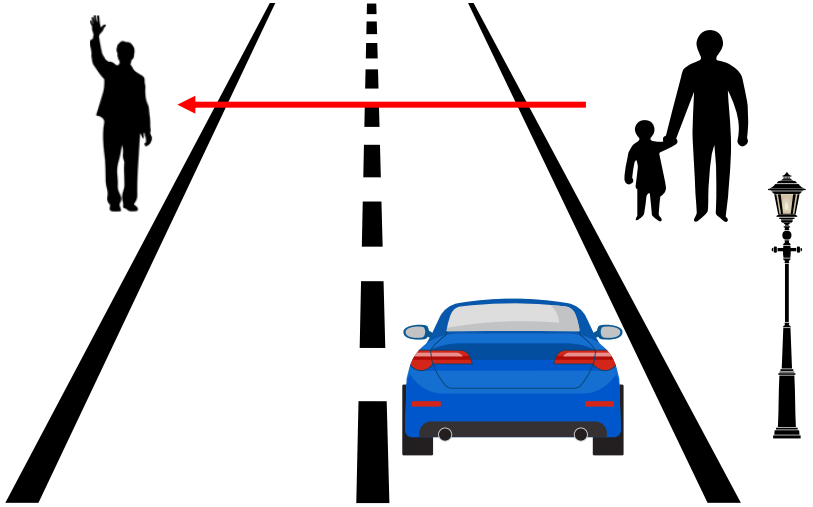
security strategies
i.e., SRE with $\varepsilon = \infty$

Definition (strategically robust equilibrium)

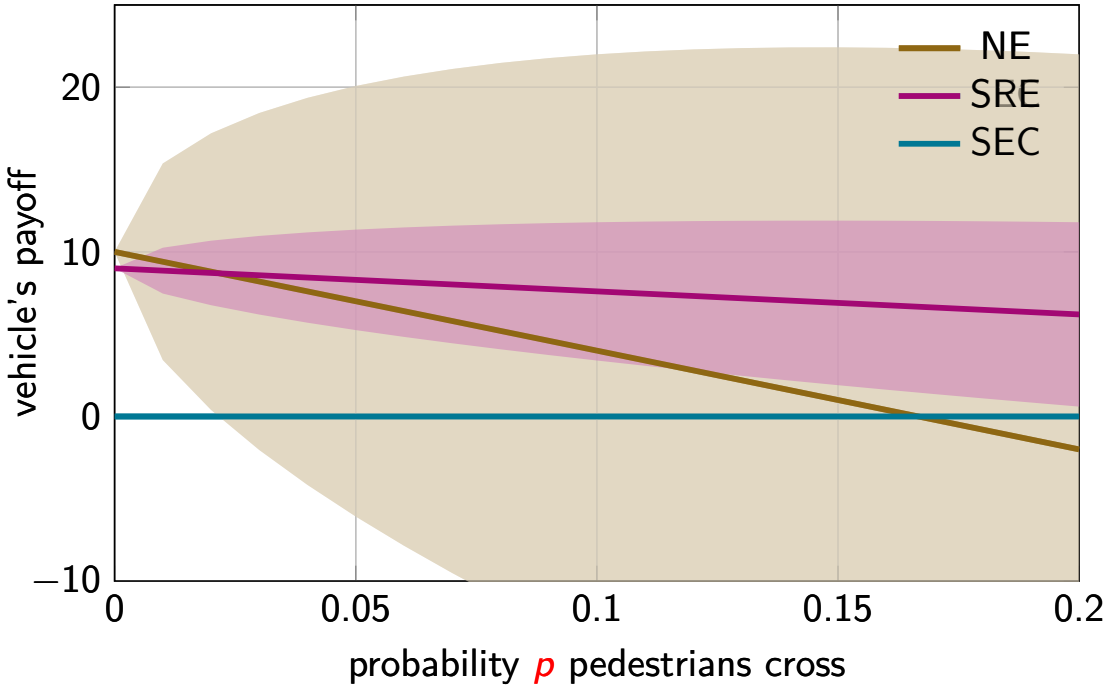
$$(\bar{p}^1, \ldots, \bar{p}^N) \quad \text{s.t.} \quad \bar{p}^i \in \operatorname*{argmax}_{p^i \in \Delta^i} \min_{\sigma^{-i} \in \mathbb{B}^i_\varepsilon(\bar{p}^{-i})} U^i(p^i, \sigma^{-i})$$

Challenges:

1) decision-dependent worst-case optimization over mixed strategies
2) fixed point of optimization problems over mixed strategies



	stop	cross
maintain	(10, –1)	(–50, –100)
decelerate	(9, –1)	(–5, –10)
stop	(0, –1)	(0, 10)



how to shape ambiguity sets?



expressivity

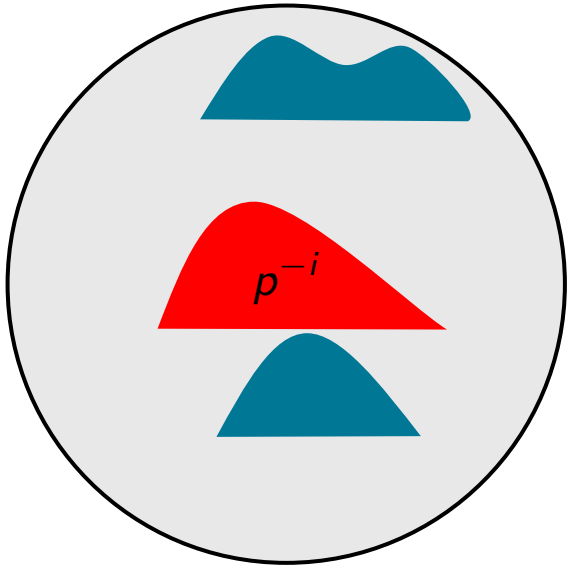


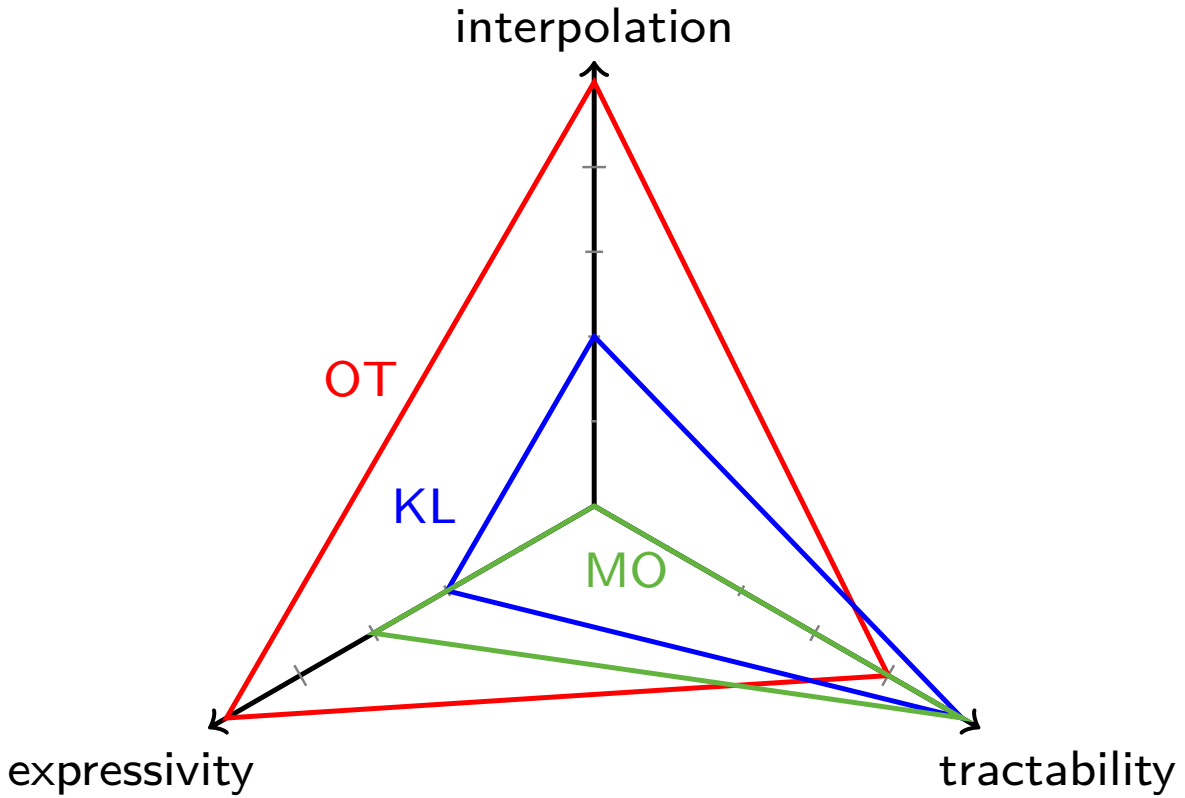
tractability



interpolation

possible approaches: optimal transport, KL-divergence, moment based, . . .





$$(\bar{p}^1, \dots, \bar{p}^N) \quad \text{s.t.} \quad \bar{p}^i \in \operatorname*{argmax}_{p^i \in \Delta^i} \min_{\sigma^{-i} \in \mathbb{B}^i_\varepsilon(\bar{p}^{-i})} U^i(p^i, \sigma^{-i})$$

Theorem (existence)

Assume that either one of the following assumptions holds true:

- Finite actions: the action space is finite;
- Continuous actions: the action space is compact and the payoff continuous.

Then, for all $\varepsilon \geq 0$, a strategically robust equilibrium exists.

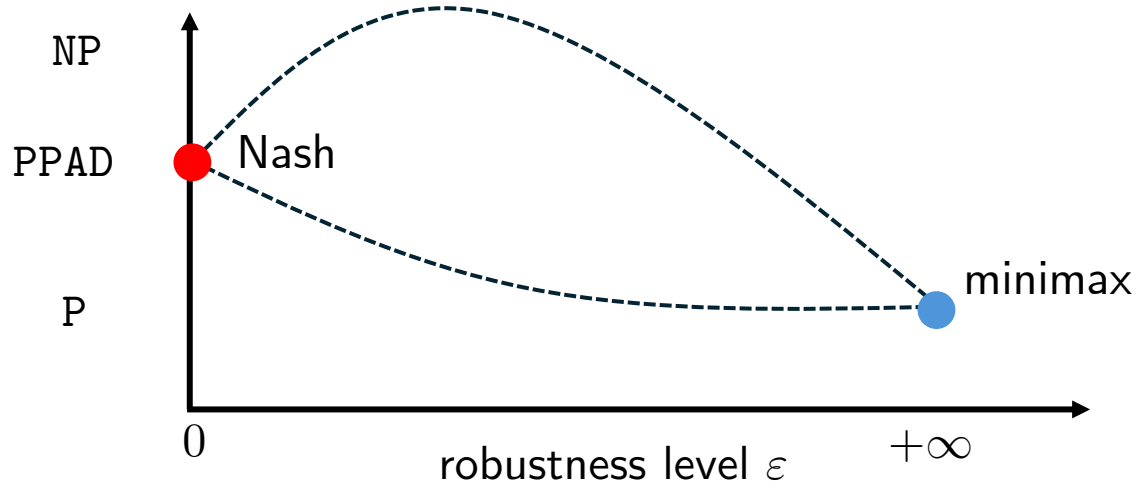
same assumptions for the existence of Nash equilibria!

Proof

- Nash's proof does not extend
- hemi-continuity of the ambiguity set
- Berge's maximum theorem
- fixed point theorem

finite games

$$(\bar{p}^1, \dots, \bar{p}^N) \quad \text{s.t.} \quad \bar{p}^i \in \underset{p^i \in \Delta^i}{\operatorname{argmax}} \min_{\sigma^{-i} \in \mathbb{B}^i_\varepsilon(\bar{p}^{-i})} U^i(p^i, \sigma^{-i})$$



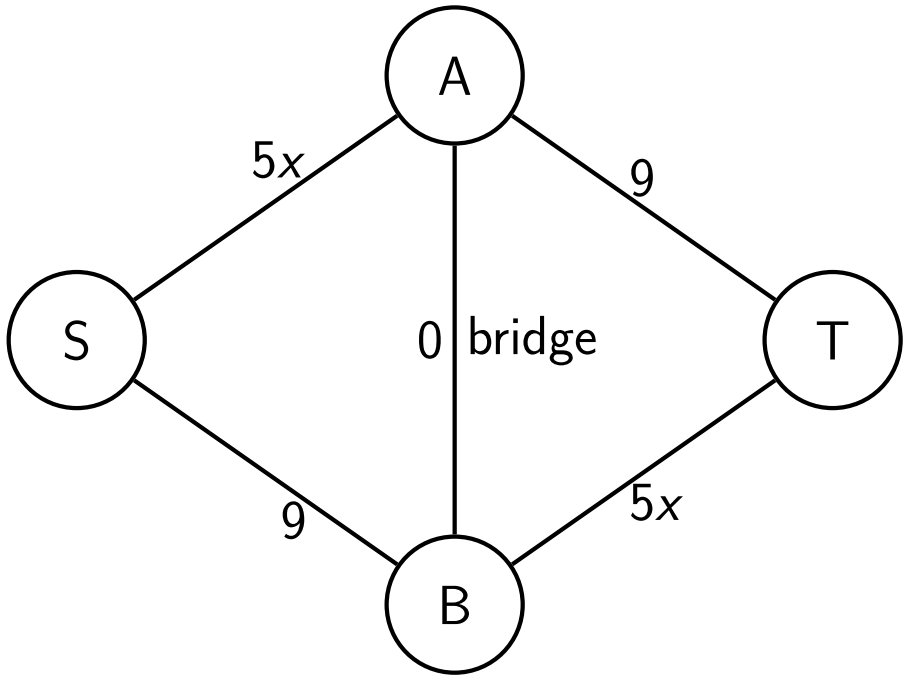
Theorem (computation in finite games)

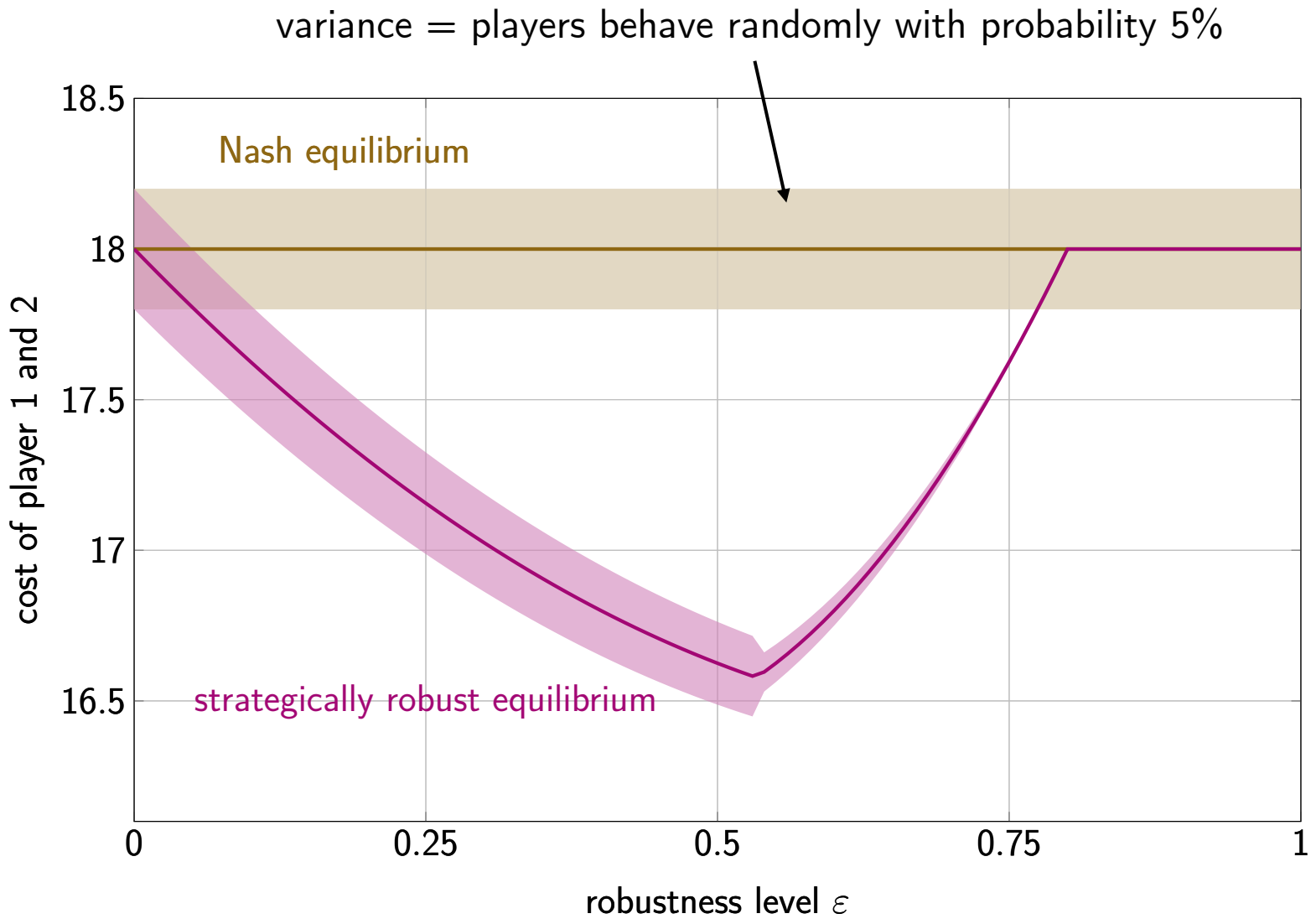
- The computational complexity of strategically robust equilibria is in PPAD.
- Strategically robust equilibria can be found by solving a multilinear complementarity problem.

Just like mixed Nash equilibria... despite the extra robustness!

 if judiciously combined with payoff robustness, then the computation is even polynomial

taming the Breass paradox





continuous games

$$(\bar{p}^1, \dots, \bar{p}^N) \quad \text{s.t.} \quad \bar{p}^i \in \underset{p^i \in \Delta^i}{\operatorname{argmax}} \min_{\sigma^{-i} \in \mathbb{B}^i_\varepsilon(\bar{p}^{-i})} U^i(p^i, \sigma^{-i})$$

Challenge: we are infinite dimensional "twice"

1) the space of mixed strategies is infinite dimensional
2) solving the worst-case optimization problem is infinite dimensional

Theorem (continuous games)

Consider a continuous concave game. Then,

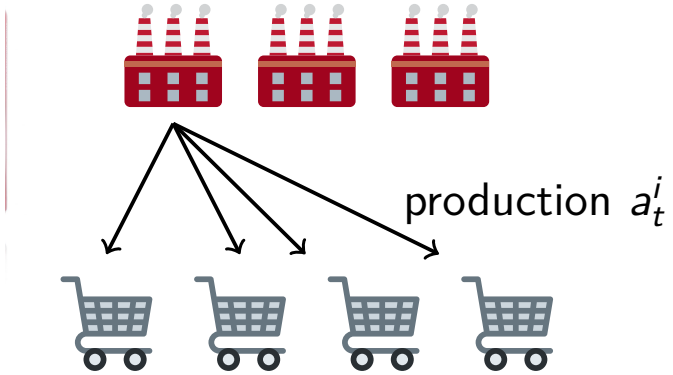
- There exists a pure strategically robust equilibrium.
- A strategically robust equilibrium is a standard Nash equilibria of a surrogate concave game, where the payoff of each agent is

$$\tilde{u}_\varepsilon((\lambda^i, a^i), (\lambda^{-i}, a^{-i})) = \min_{\tilde{a}^{-i} \in \mathcal{A}^{-i}} u^i(a^i, \tilde{a}^{-i}) - \lambda^i d^i(a^{-i}, \tilde{a}^{-i}) - \lambda^i \varepsilon$$

no infinite dimensionality + tools "à la Nash for computation"



N firms compete in T markets

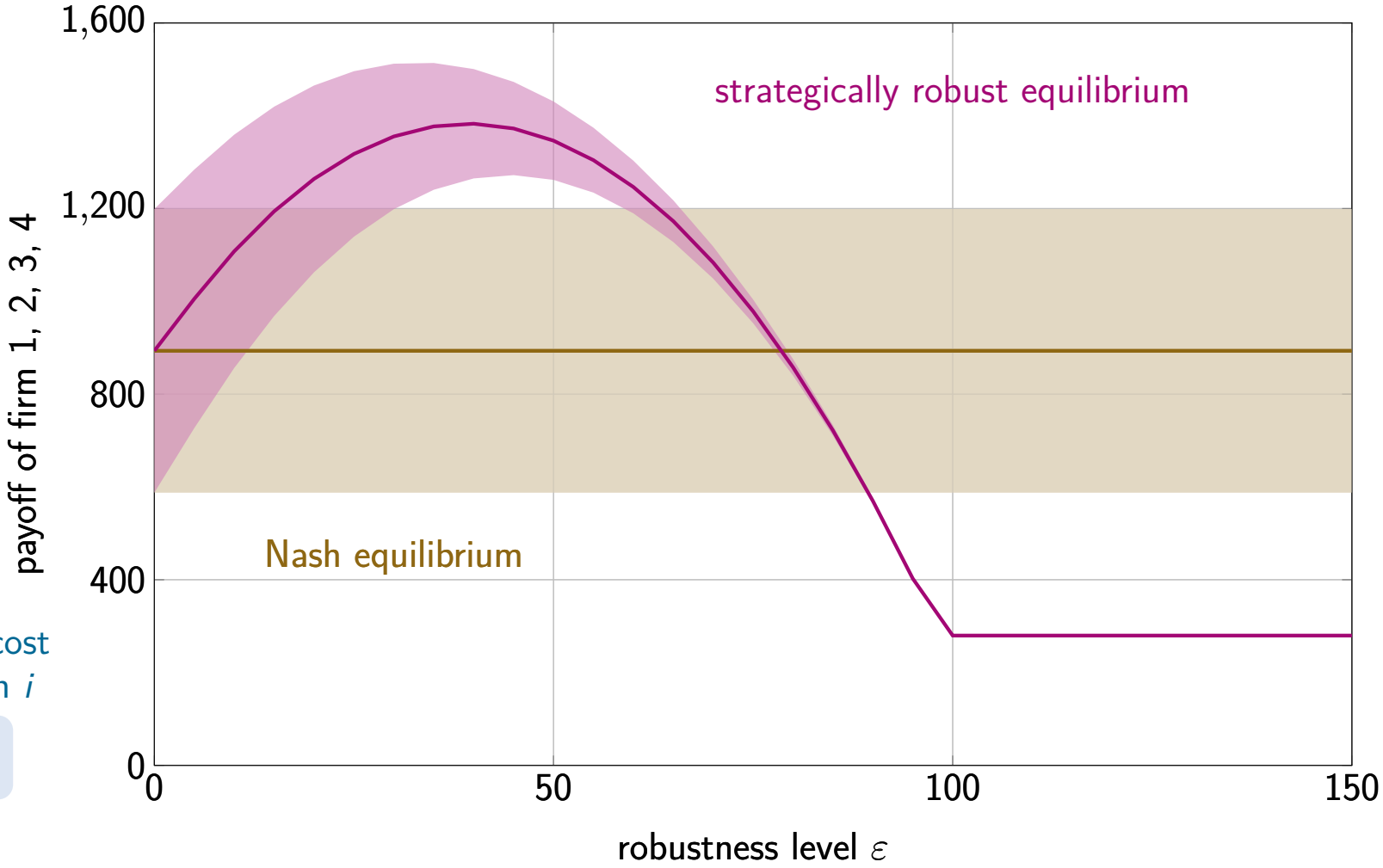


$$\text{profit}^i(a^i, a^{-i}) = \sum_{t=1}^{T} a_t^i \left(\alpha_t - \beta_t \sum_{i=1}^{N} a_t^i \right) - c^i a_t^i$$

price in market t

prod. cost of firm i

production constraints: $\sum_t a_t^i \in [0, K^i]$

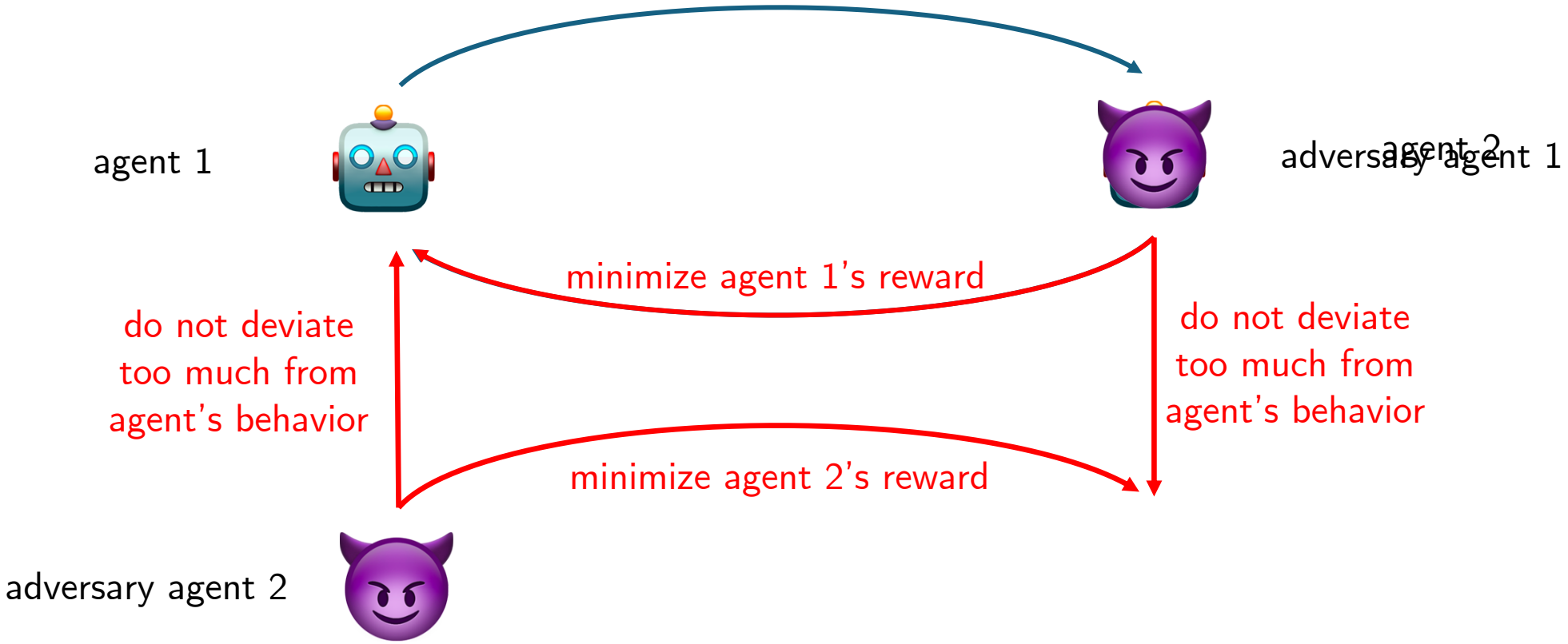


from game-theoretic models to AI agents

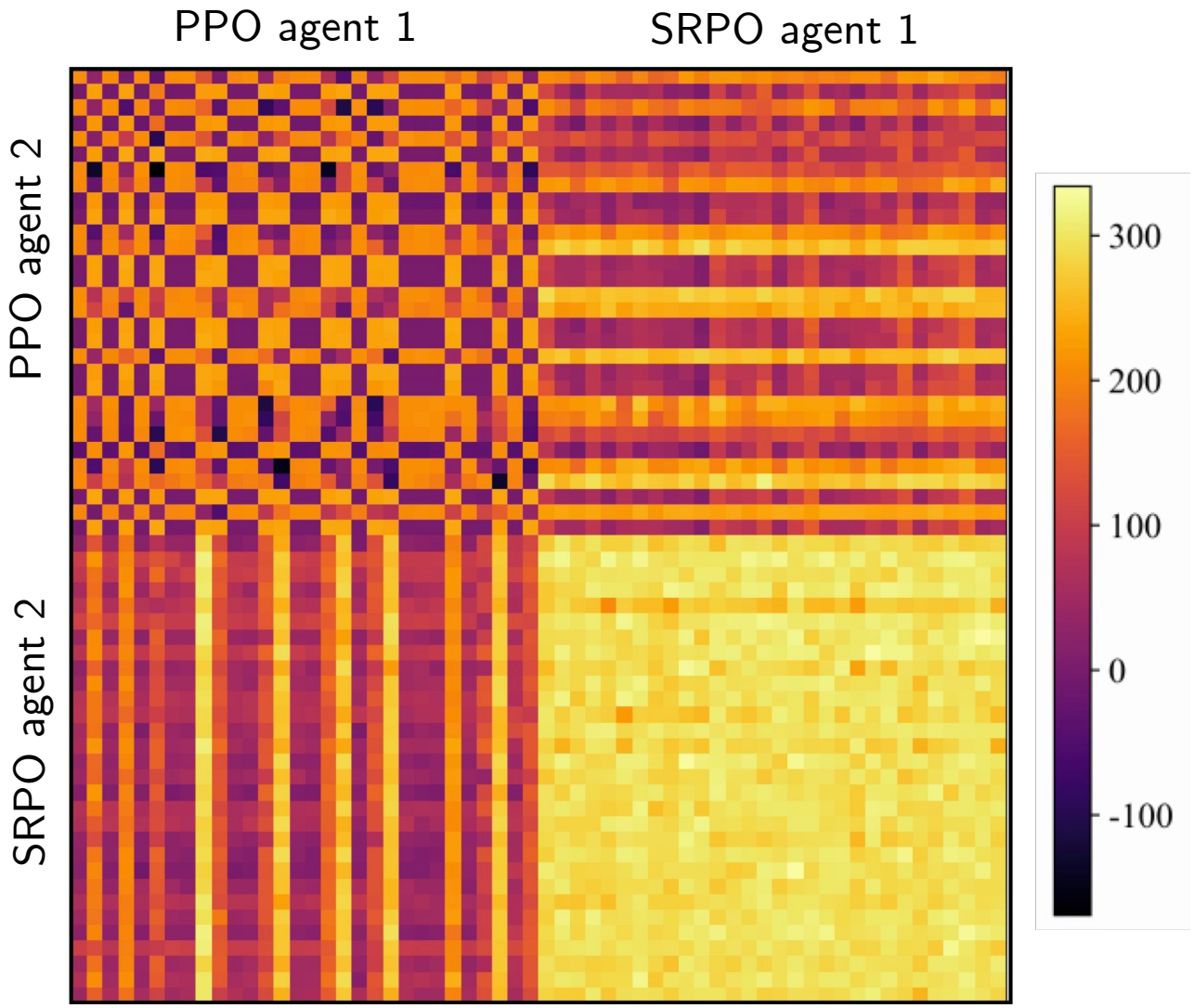
with C. Qu, Y. Zhang, E. Mazumdar

$$(\bar{p}^1, \ldots, \bar{p}^N) \quad \text{s.t.} \quad \bar{p}^i \in \underset{p^i \in \Delta^i}{\text{argmax}} \; \min_{\sigma^{-i} \in \mathbb{B}^i_\varepsilon(\bar{p}^{-i})} U^i(p^i, \sigma^{-i})$$

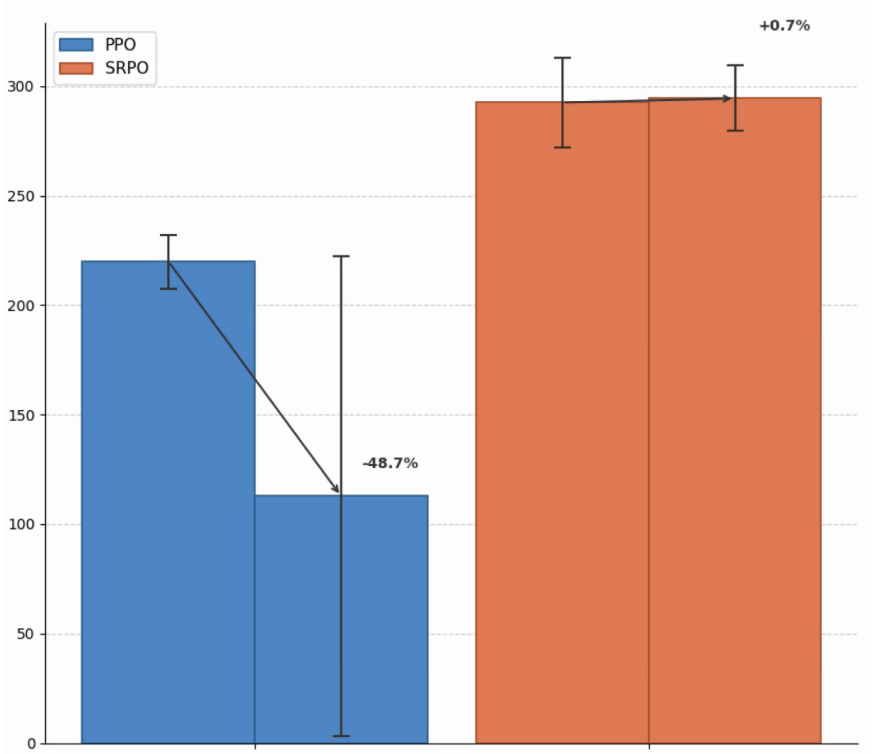
key insight: train agent against a fictitious adversary "regularized around" the other player



back to the AI lab



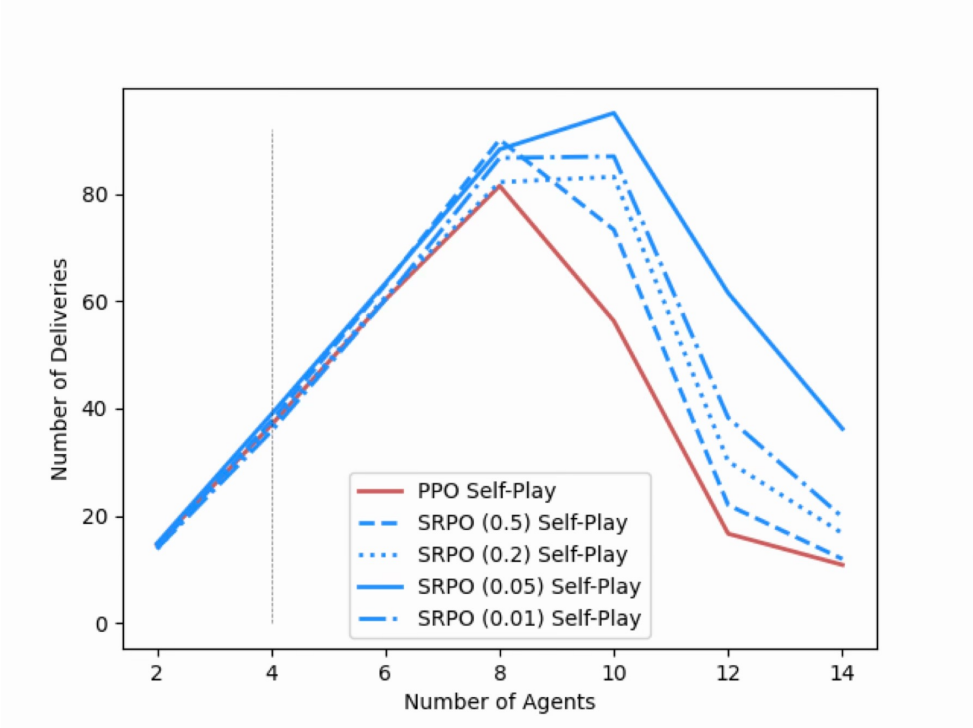


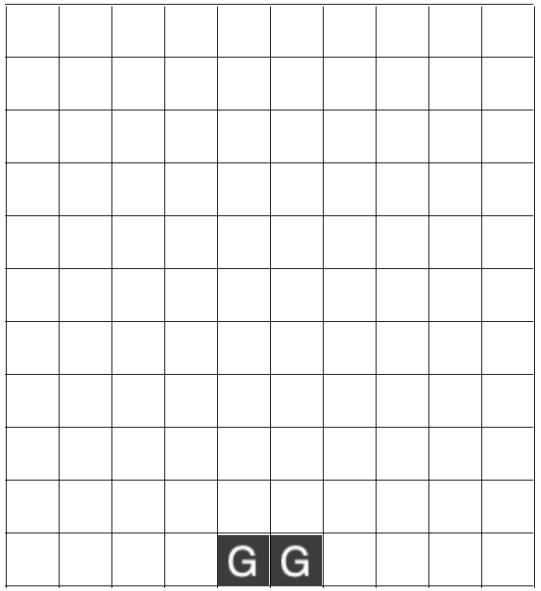


warehouse

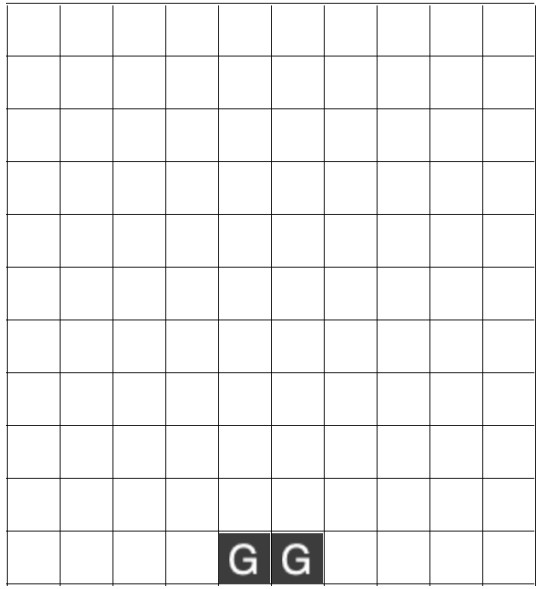
Agents have limited observations
(3x3 grid centered at their current state)

+1 reward for delivering shelf to the goal
+0.01 reward for picking up shelf

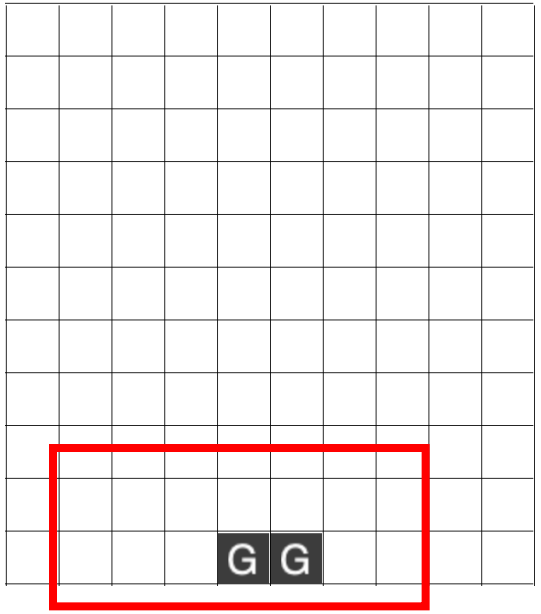


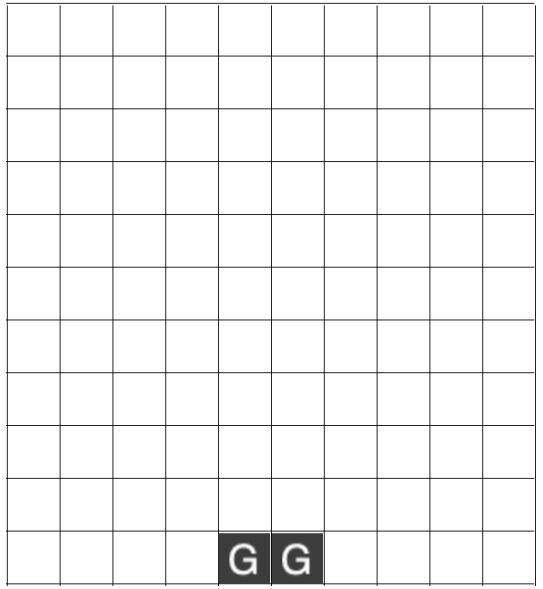


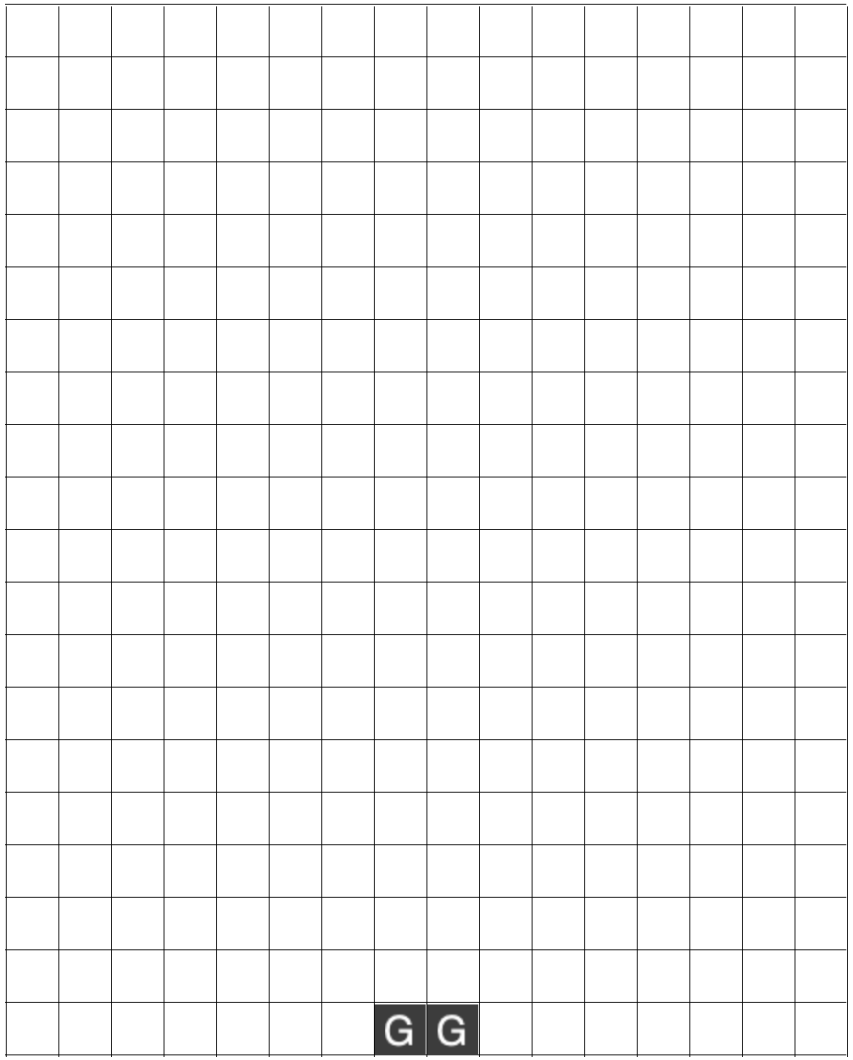
PPO



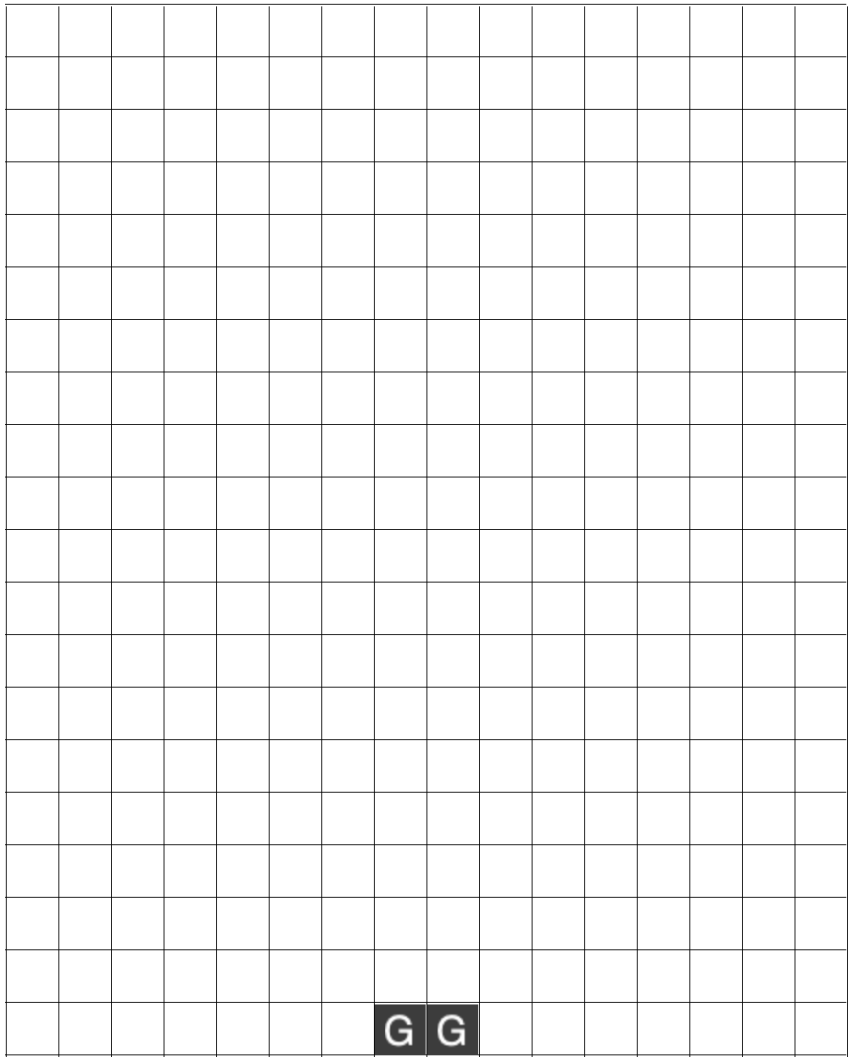
SRPO (ours)







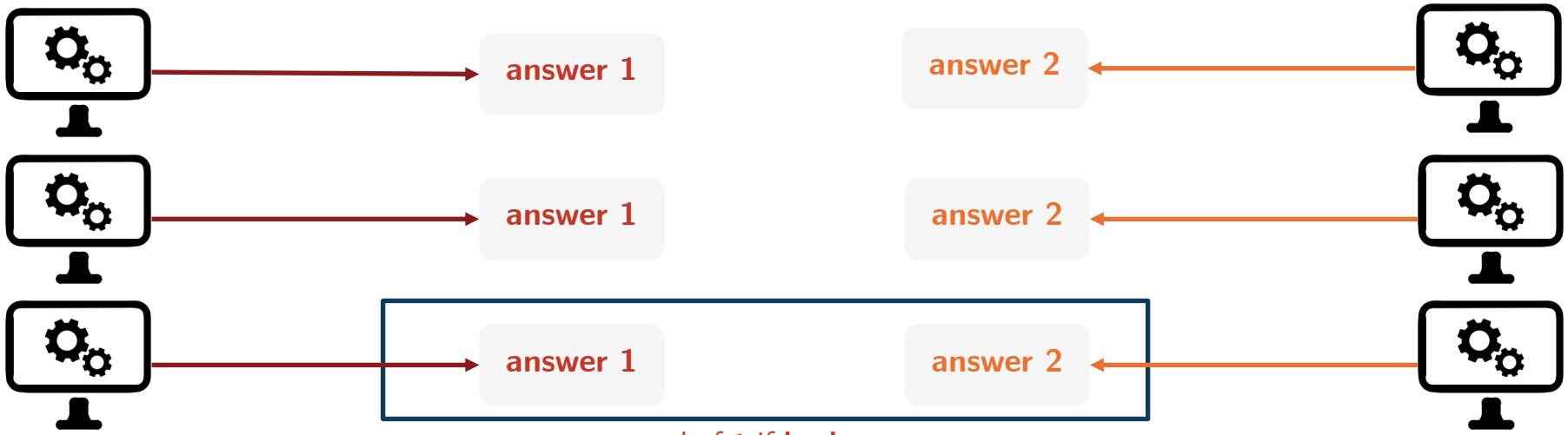
PPO



SRPO (ours)

multi-LLM-agent debate on GSM8K

Question: Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?



reward of 1 if **both** are correct.





Method	Q0.5B&Q0.6B	Q0.5B&Q3B	Q0.5B&Q4B	Q0.6B&Q3B	Q0.6B&Q4B	Q3B&Q4B
SRPO	**0.622**	**0.686**	**0.873**	**0.616**	**0.848**	**0.732**
IPPO	0.605	0.651	0.846	0.573	0.711	0.709
Improvement (%)	+2.81%	+5.38%	+3.19%	+7.50%	+19.27%	+3.24%

untuned

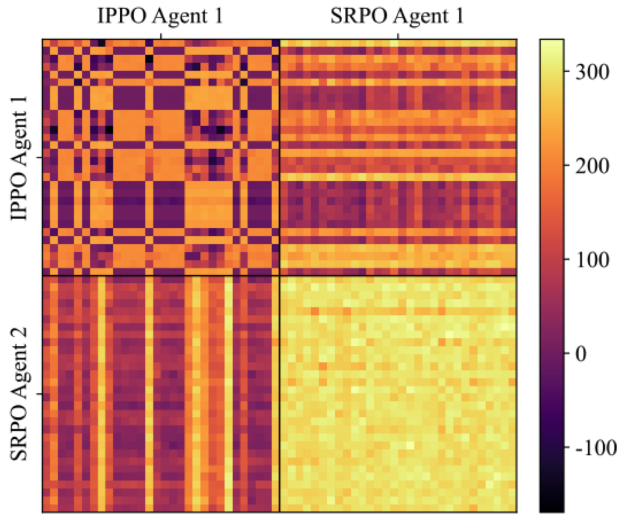




Method	Q0.5B	Q0.6B	Q3B	Q4B
SRPO	**0.405**	**0.671**	**0.632**	**0.917**
IPPO	0.378	0.587	0.552	0.901
Improvement (%)	+7.14%	+14.31%	+14.49%	+1.78%

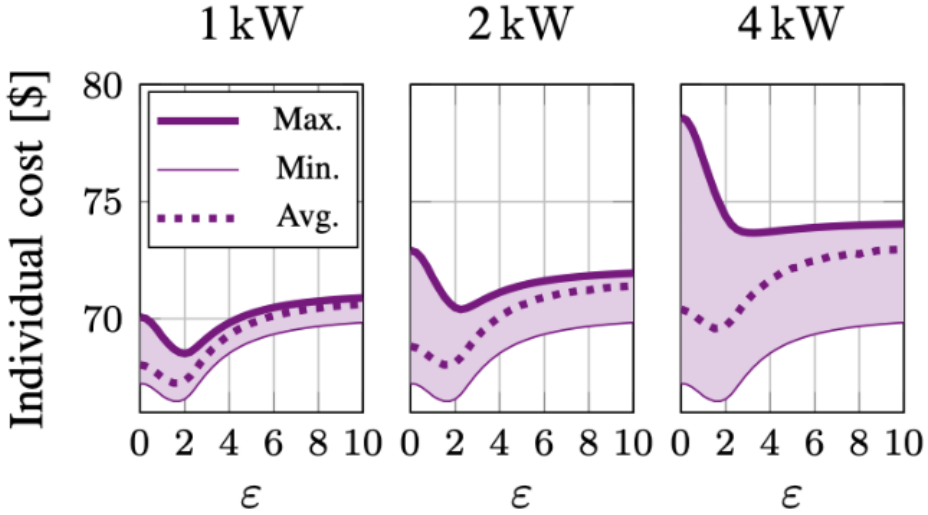
training collaborative AI agents

(with C. Qu, Y. Zhang, E. Mazumdar)



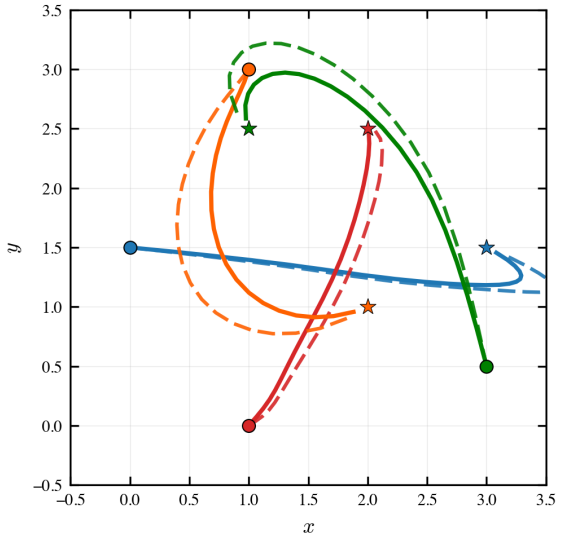
electric vehicle charging

(with A. Feik, S. Bolognani, F. Dörfler, D. Paccagnan)



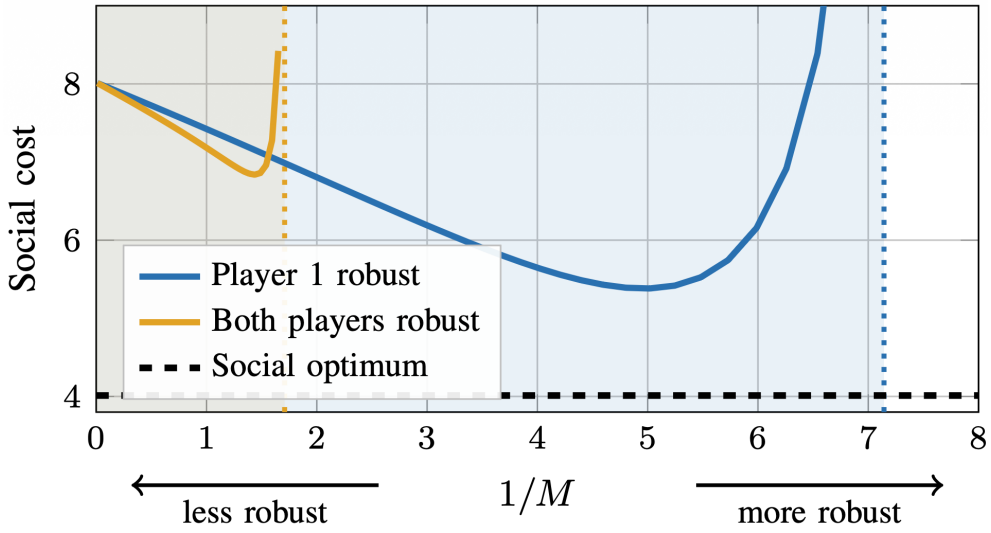
strategically robust trajectory generation

(with V. Qin, S. Bolognani, H. Balakrishnan)



strategically robust dynamic games

(with B. Velasevic and E. Mazumdar)



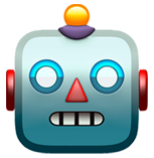
we developed strategically robust decision making that



... protects agents against deviations in other's behavior



... often yields higher payoffs for all agents



... leads to more cooperative and generalizable AI agents



... comes at no additional cost compared to NE, despite the robustness